

Quantum Information

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Abstract

Extremely rough notes on quantum information. Maybe these will turn into something coherent, maybe not!

Contents

1 POVM Measurements	1
1.1 Pure States and Projective Measurements	1
1.2 Entangled States and POVMs	2
2 Pure State Estimation, Symmetric Subspace	3
2.1 Continuous POVMs and the Uniform Measure	3

1 POVM Measurements

1.1 Pure States and Projective Measurements

As typically introduced, the quantum mechanics of pure states $|\psi\rangle$ living in a Hilbert space $\mathcal{H}$ connects to experiments via projective measurements. Formally, a projective measurement is described by an indexed collection of projectors

$$\{\Pi_x : \mathcal{H} \rightarrow \mathcal{H} \mid x \in S\}, \quad S \subset \mathbb{R} \quad (1)$$

which add up¹ to the identity

$$\sum_{x \in S} \Pi_x = \mathbf{1} \quad (2)$$

so that they equivalently specify an internal direct sum decomposition

$$\mathcal{H} = \bigoplus_{x \in S} \mathcal{H}_x, \quad \mathcal{H}_x \equiv \Pi_x(\mathcal{H}). \quad (3)$$

A projective measurement then randomly projects any prepared pure state $|\psi\rangle \in \mathcal{H}$

$$|\psi\rangle \xrightarrow{\text{proj}} \frac{\Pi_x |\psi\rangle}{\|\Pi_x |\psi\rangle\|} = \frac{\Pi_x |\psi\rangle}{\sqrt{\langle\psi|\Pi_x|\psi\rangle}} \quad (4)$$

and the corresponding $x \in S$ is called the measurement outcome. Assuming WLOG that $|\psi\rangle$ is properly normalized $\langle\psi|\psi\rangle = 1$, the probability of any particular outcome is given by the Born rule

$$\text{Prob}_x = \|\Pi_x |\psi\rangle\|^2 = \langle\psi|\Pi_x|\psi\rangle. \quad (5)$$

If $S \subset \mathbb{R}$ is anything more analytically interesting than a finite set, this object should be interpreted with care as a probability measure. For the present discussion, it suffices to imagine that S is set.

¹Here “addition” is meant generously to include integration.

1.2 Entangled States and POVMs

The above is all well and good, but the following simple and natural thought experiment suggests a more general kind of process which does not quite take the form of a projective measurement, but which intuitively warrants being called a “measurement” of our system. Suppose we are interested in the physics of some Hilbert space $\mathcal{H}_A$, but that there exists a second space $\mathcal{H}_B$ and that we have the experimental power to perform general joint measurements. A general joint state

$$|\Psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B \quad (6)$$

is entangled, but for our purposes it will suffice to consider a pure state

$$|\Psi\rangle_{AB} = |\psi\rangle \otimes |\phi_{\text{aux}}\rangle \quad (7)$$

where we think of the second factor as an auxiliary state. Now suppose we perform a projective measurement on the joint state, described as above by a set of projectors

$$\{\Pi_{AB,x} \mid x \in S \subset \mathbb{R}\}. \quad (8)$$

The Born rule applied to the joint state gives the probability of outcome $x \in S$ as

$$\text{Prob}_x = (\langle\psi| \otimes \langle\phi_{\text{aux}}|) \Pi_{AB,x} (|\psi\rangle \otimes |\phi_{\text{aux}}\rangle). \quad (9)$$

If we think of this as a function of $|\psi\rangle$ at fixed $|\phi_{\text{aux}}\rangle$ and $\Pi_{AB,x}$, we can naturally rewrite the result as the diagonal matrix element of a quadratic form Q_x on $\mathcal{H}$ with matrix elements

$$Q_x : \mathcal{X}_A \times \mathcal{H}_A \rightarrow \mathbb{C}, \quad \langle\phi| Q_x |\chi\rangle \equiv (\langle\phi| \otimes \langle\phi_{\text{aux}}|) \Pi_{AB,x} (|\chi\rangle \otimes |\phi_{\text{aux}}\rangle). \quad (10)$$

The RHS is antilinear in the first argument and linear in the second, so this defines a bona fide quadratic form, and we can identify

$$\text{Prob}_x = (\langle\psi| \otimes \langle\phi_{\text{aux}}|) \Pi_{AB,x} (|\psi\rangle \otimes |\phi_{\text{aux}}\rangle) = \langle\psi| Q_x |\psi\rangle. \quad (11)$$

Recall that the polarization identity on quadratic forms

$$Q(u, v) = \frac{Q(u + v, u + v) - Q(u, u) - Q(v, v)}{2} \quad (12)$$

tells us a quadratic form’s entire action is fixed by its diagonal values, so knowing the value of $\langle\psi| Q_x |\psi\rangle$ for all $|\psi\rangle \in \mathcal{H}_A$ (or more weakly by linearity, just its values on unit-normalized such states) fixes Q_x uniquely.

The defining properties of $\Pi_{AX,b}$ as a projective measurement, plus judicious use of the polarization identity, imply that each Q_x is a Hermitian positive semi-definite operator²

$$Q_x = Q_x^\dagger, \quad Q_x \geq 0 \quad (13)$$

and that the collection taken together is a decomposition of the identity

$$\sum_{x \in S} Q_x = \mathbf{1}_A. \quad (14)$$

These two properties define a *positive operator-valued measurement* (POVM). The reader can convince themselves that any such collection $\{Q_x\}_{x \in S}$ with these two properties can be cooked up as we did here, by a suitable choice of auxiliary state $|\phi_{\text{aux}}\rangle \in \mathcal{H}_B$ and joint projective measurement $\{\Pi_{AB,x}\}_{x \in S}$.

²Here we are slightly redundant, as positive semi-definiteness is only defined for Hermitian operators to begin with.

2 Pure State Estimation, Symmetric Subspace

In the words of Michael Walter, we now address the

Goal (Pure State Estimation): Find a POVM $\{Q_{\hat{\phi}}\}_{\hat{\phi} \in \Omega}$ on $(\mathbb{C}^d)^{\otimes n}$, with possible outcomes in the set

$$\Omega = \left\{ \rho_{\hat{\phi}} = |\hat{\phi}\rangle\langle\hat{\phi}| \right\} \quad (15)$$

of pure states on $\mathbb{C}^d$, such that when we measure $\phi^{\otimes n} = |\phi\rangle^{\otimes n}$, we obtain an outcome $\hat{\phi}$ that is “close” to ϕ .

As a small clarification: when we talk about outcomes, we are not saying that the POVM projects down to pure states on $\mathbb{C}^d$. Indeed, in general a POVM does not admit an interpretation as a projector with a pure state measurement outcome. We mean that we have some collection of pure states in $\mathbb{C}^d$; that we are using this collection as our indexing set for the POVM; and that when our measurement returns the outcome associated with $|\hat{\phi}\rangle$, we interpret this as our best guess for the state whose n -fold product is our initial pre-measurement state. A well-designed such POVM at variable n should have the property that, in the limit of large n , this procedure should yield guesses $|\hat{\phi}\rangle$ which are extremely similar to $|\phi\rangle$ extremely often. This is plausible essentially because, given two unit-normalized states $|\psi\rangle, |\xi\rangle \in \mathbb{C}^d$, we have

$$|\langle\psi|\xi\rangle| < 1 \implies \left| \left(\langle\psi|^{\otimes n} \right) \left(|\xi\rangle^{\otimes n} \right) \right| = |\langle\psi|\xi\rangle|^n \xrightarrow{n \rightarrow \infty} 0. \quad (16)$$

So if two states are at all distinguishable, then their corresponding n -fold copies are very distinguishable for large enough n . Of course the order of predicates is important. This is a pointwise limit (for any two fixed distinct states, distinguishability goes to 1 as n goes to ∞), not a uniform one (to the contrary, given any n no matter how large, we could adversarially produce distinct states whose n -fold products are arbitrarily hard to distinguish).

2.1 Continuous POVMs and the Uniform Measure

In order to get a typical guess $|\hat{\phi}\rangle$ which is arbitrarily close to the true state $|\phi\rangle$ at large n no matter what the latter happens to be, our indexing set $\Omega \subset \mathbb{C}^d$ must at least contain states arbitrarily close to any possible true state. Rather than taking Ω itself to depend on n , we will take Ω from the state to be uncountably infinite, covering all distinct states in $\mathbb{C}^d$. As two states are distinct if they correspond to different rays, this means our indexing set is $\Omega \simeq \mathbb{CP}^{d-1}$.

Now the correct probabilistic interpretation of our outcomes must be more explicitly measure-theoretic. A continuous POVM is specified by an indexed collection of operators on $\mathcal{H}$

$$\{Q_x : \mathcal{H} \rightarrow \mathcal{H}\}_{x \in \Omega} \quad (17)$$

plus the auxiliary data of a measure dx on Ω . We require as before that each operator be positive semi-definite and that the collection jointly be a decomposition of the identity

$$Q_x \geq 0 \ \forall x \in \Omega, \text{ and } \int_{\Omega} dx Q_x = \mathbf{1}. \quad (18)$$

For any $\psi \in \mathbb{C}^d$ and $x \in \Omega$, the probability density of outcomes

$$p_{\psi}(x) = \langle\psi| Q_x |\psi\rangle \quad (19)$$

gives rise to finite probabilities

$$\text{Prob}_{\psi}(\text{outcome} \in S) = \int_S dx \langle\psi| Q_x |\psi\rangle \text{ for } S \subset \Omega \text{ measurable} \quad (20)$$

and expected values

$$\mathbb{E}_\psi[f : \Omega \rightarrow \mathbb{C}] = \int_\Omega dx \langle \psi | Q_x | \psi \rangle f(x) \quad (21)$$

So that all of the above makes analytic sense, the mapping $x \mapsto Q_x$ must be measurable. Notice then that the mapping Q which takes measurable subsets $X \subset \Omega$ to operators

$$Q(X) := \int_X dx Q_x \quad (22)$$

can itself be thought of as a measure on Ω taking positive(-semidefinite)-operator values, justifying the term POVM.

For the particular problem of pure state estimation, even before we decide how to map states $|\hat{\phi}\rangle$ to operators $Q_{\hat{\phi}}$, we must decide on an appropriate measure. Conjugation by any fixed unitary as a symmetry of all observable physics, so any physically sensible measure must respect this symmetry.

$$\int d\psi f(\psi) = \int d\psi f(U\psi U^\dagger). \quad (23)$$

Just as there is a unique rotation-invariant measure on the sphere up to overall scaling, this symmetry uniquely fixed $d\psi$ to be the *Haar measure* after we fix scaling by demanding that $d\psi$ be a probability measure.